sopvar Scaling Benchmarks

Time and storage of the standard class operations, and parsing time of the stability and output feedback LPIs, for the operator classes sopvar and sdopvar and the classes they replace, opvar and dopvar.

measured 11/05–11/06/2026 ndopvar @ edbd7341, 8343fd1b, 392a6adf MATLAB R2025b · MOSEK 11.0.30
Intel i9-14900KF · 63.7 GB

SUMMARY

On constant operators, sopvar takes 0.03–0.58 of the opvar time on every standard operation except the adjoint, whose ratio reaches 2.16 at $n = d = 8$. On operators linear in q decision variables with a fixed number of nonzero decision coefficients, every sdopvar operation with one decision operand stays flat in q up to 10^6, while the dopvar composition grows linearly: 2.95 s against 2.9 ms at $q = 10^6$. Adding two decision operators costs time growing with q in both classes; sdopvar takes 2.3 times the dopvar time when the decision-variable lists are disjoint and 0.03 of it when they are shared. Parsing the 1-D stability LPI takes 0.08–0.15 of the legacy time for the same SDP – 2.5 s against 16.4 s at Lyapunov degree 12 – which lowers the share of total time spent parsing from 64% to 19%. In 2-D at Lyapunov degree 4, measured without the solve, the container parses the stability LPI in 16.6 s against 56.4 s. The H∞ output feedback LPI, which has no legacy implementation, reaches 14.6 million decision variables at degree 4 in 2-D – 4.8 times the stability LPI at that degree – and parses in 92 s.

1 Setup

Classes opvar and sopvar represent constant PI operators; dopvar and sdopvar represent operators linear in decision variables $\xi \in \mathbb{R}^q$, the unknowns of an LPI. opvar and dopvar store each kernel as a polynomial object (multipoly, dpvar). sopvar and sdopvar store coefficient matrices in monomial bases, $C(\xi) = A + B^{\mathrm{T}}\xi$ for the decision classes.

Operators Every test operator acts on $L_2{}^n[0,1]$, with R_0, R_1, R_2 polynomial of degree d in s and in t:

$$(Px)(s) = R_0(s)\,x(s) + \int_0^s R_1(s,t)\,x(t)\,dt + \int_s^1 R_2(s,t)\,x(t)\,dt.$$

Equivalence Each operator is built once in the legacy class and converted once (`opvar2sopvar`, `dopvar2sdopvar`), so both classes hold the same operator. A check applies every operation in both classes, then applies both results to 3 random quadratic inputs at 60 points, each class from its own definition; decision results are first evaluated at random values of ξ. Over 22 operations the largest relative pointwise difference is $3.7\cdot10^{-16}$.

Timing One warm-up call, then the median of 3–25 calls filling at least 0.3 s. For an LPI, parsing time is the wall time of the executive minus the solver time reported by MOSEK, the definition used in Jagt et al. (2022). A run with a stub in place of MOSEK measures parsing alone (Section 5).

● opvar / dopvar (legacy) ▼ sopvar / sdopvar ■ sopvar / sdopvar on the stock SDP or slack
○ open marker: parse only, no solve

2 Operations at fixed size

Constant operators on a grid $n, d \in \{1, 2, 4, 8\}$. Over the whole grid sopvar takes 0.03–0.58 of the opvar time on every operation except the adjoint. For $n, d \le 8$ every operation except composition costs a fixed amount per call: the least-squares exponents of time in n and in d stay within ±0.17 in both classes, so these times do not extrapolate to larger sizes. Composition grows in both classes and shows the largest gap. The sopvar adjoint grows with size (exponent 0.17 in n, 0.15 in d) where the opvar adjoint does not, so the ratio rises from 0.50 to 2.16.

Constant operators. Times at $n = d = 8$; ratio = sopvar / opvar.

OPERATION	OPVAR (S)	SOPVAR (S)	RATIO AT 8, 8	RATIO OVER GRID
A + B	1.61e-3	5.05e-4	0.31	0.24 – 0.58
A − B	1.93e-3	4.69e-4	0.24	0.24 – 0.45
−A	1.45e-4	1.26e-5	0.09	0.04 – 0.09
2.5·A	2.59e-4	2.48e-5	0.10	0.07 – 0.10
A·B (composition)	0.252	0.0266	0.11	0.03 – 0.11
A′ (adjoint)	4.99e-4	1.08e-3	2.16	0.50 – 2.16
[A B]	1.72e-3	4.80e-4	0.28	0.28 – 0.42
[A; B]	1.66e-3	4.64e-4	0.28	0.27 – 0.42
blkdiag(A, B)	5.03e-3	5.27e-4	0.10	0.10 – 0.12
A(1:n/2, :)	9.79e-4	2.40e-4	0.25	0.25 – 0.39
INVERSE OF I + 0.1K, N ≤ 4, D ≤ 2				
A^{-1}	0.10 – 0.32	0.017 – 0.058	0.12 – 0.23	residual ≤ 1.7e-10 vs 1.5e-9

3 Composition against the number of decision variables

The analogue of Fig. 2 in Jagt et al. (2022): scalar operators ($n = 1$) with kernels of degree 4, so 55 coefficients. In the left panel each coefficient carries one decision variable drawn from q, which holds the number of nonzero decision coefficients at nnz = 110 as q grows from 10^3 to 10^6. In the right panel each coefficient carries $[4q/55]$ variables, so nnz ≈ $3.9q$.

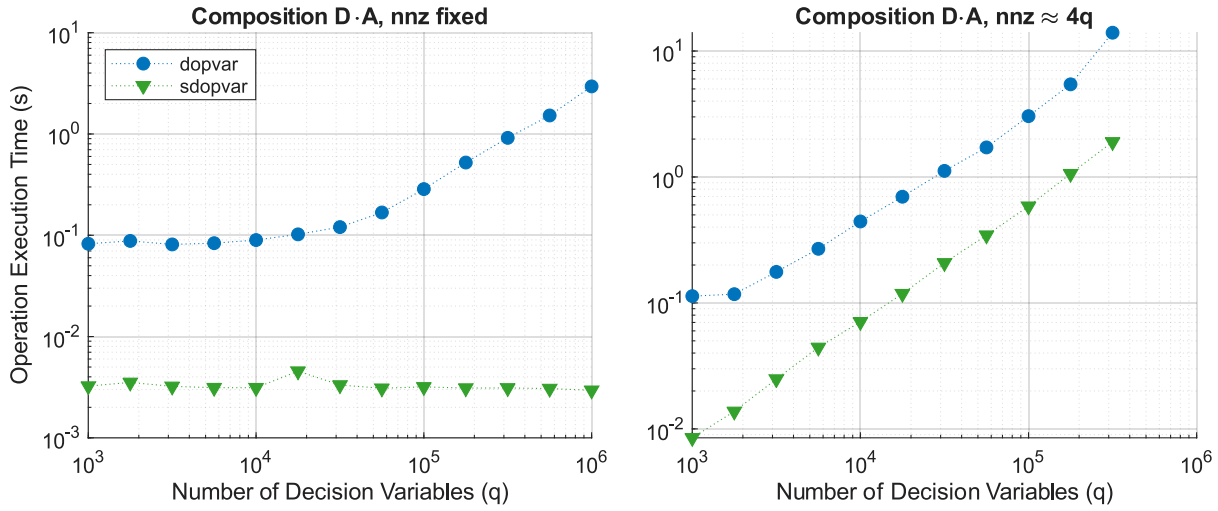


Figure 1. Time of the composition $D{\cdot}A$ of a decision operator D with a constant operator A against the number of decision variables q. Left: nnz fixed at 110. Right: nnz ≈ $3.9q$, up to $q = 3.2{\cdot}10^5$.

With nnz fixed, the sdopvar composition $D{\cdot}A$ takes 2.9–4.6 ms at every q (q-exponent −0.03 over the last decade), while the dopvar composition grows from 0.082 s to 2.95 s (q-exponent 1.01). With nnz $\propto q$ both grow with nnz – sdopvar with q-exponent 0.96, dopvar with 1.10 – and sdopvar takes 0.14 of the dopvar time at $q = 3.2{\cdot}10^5$ (1.90 s against 13.98 s).

4 Addition and adjoint against the number of decision variables

The analogue of Fig. 3 in Jagt et al. (2022). The decision classes have no differentiation or substitution, so the panels show addition and the adjoint; the data file holds every operation.

Addition. Adding two decision operators needs one list of their decision variables. With disjoint lists both classes merge the two lists: over the last decade the q-exponent is 1.10 for dopvar and 1.28 for sdopvar, and at $q = 10^6$ the addition takes 1.53 s and 3.58 s. With one shared list sdopvar detects the equal lists and skips the merge; its time grows as q (q-exponent 0.99) from 0.29 ms to 51 ms, while dopvar merges regardless and takes 1.75 s at $q = 10^6$. In the code, `@sdopvar/private/sync_basis` tests the lists with `isequal` and otherwise forms `unique` of their concatenation and re-indexes each operand through `ismember`; `@dpvar/plus` concatenates the lists and merges them with `combine` in every case. This account comes from reading the code; it was not profiled.

Adjoint. Flat in q in both classes, at 0.77–0.84 ms (dopvar) and 0.38–0.45 ms (sdopvar) from $q = 10^3$ to 10^6.

Every operation at $q = 10^6$, nnz = 110, with the q-exponent of time over $q = 10^5$–10^6. Green: flat in q (|exponent| < 0.1). Blue: grows with q. Each exponent comes from two points; below 1 ms that carries noise – the 0.44 of dopvar −D (shared list) rests on one 0.8 ms point.

OPERATION	DOPVAR (S)	EXP.	SDOPVAR (S)	EXP.	SHARED LIST: DOPVAR (S)	EXP.	SDOPVAR (S)	EXP.
ONE DECISION OPERAND								
−D	2.93e-4	−0.01	1.39e-5	0.02	8.11e-4	0.44	1.36e-5	0.02
2.5·D	2.25e-3	−0.07	1.46e-5	−0.06	2.17e-3	0.02	1.57e-5	0.06
D + A (constant A)	2.48	1.26	4.57e-4	0.02	2.38	1.23	3.96e-4	−0.05
D·A	2.95	1.01	2.95e-3	−0.03	2.87	1.03	2.88e-3	0.01
A·D	2.87	0.99	4.52e-3	−0.09	2.91	1.03	4.51e-3	0.03
D′	7.72e-4	−0.04	4.18e-4	0.01	7.91e-4	0.01	4.01e-4	0.03
D(1, :)	3.82e-2	1.06	1.92e-4	0.05	2.84e-2	0.94	1.63e-4	−0.03
TWO DECISION OPERANDS								
D + E	1.53	1.10	3.58	1.28	1.75	1.15	5.10e-2	0.99
D − E	1.50	1.08	3.58	1.25	1.50	1.04	5.54e-2	1.03
[D E]	3.44	1.16	3.36	1.23	3.44	1.19	5.02e-2	0.97
[D; E]	3.69	1.22	3.60	1.23	3.50	1.22	5.64e-2	1.02
blkdiag(D, E)	3.78	1.19	3.62	1.24	3.83	1.20	8.72e-2	0.97

With nnz $\approx 3.9q$ (q = $3.2{\cdot}10^4$–$3.2{\cdot}10^5$) the compositions grow with q-exponent 0.95–0.96 (sdopvar) and 1.08–1.10 (dopvar), and the two-operand operations with 1.24–1.40 in both classes; D + A remains flat in sdopvar (−0.01).

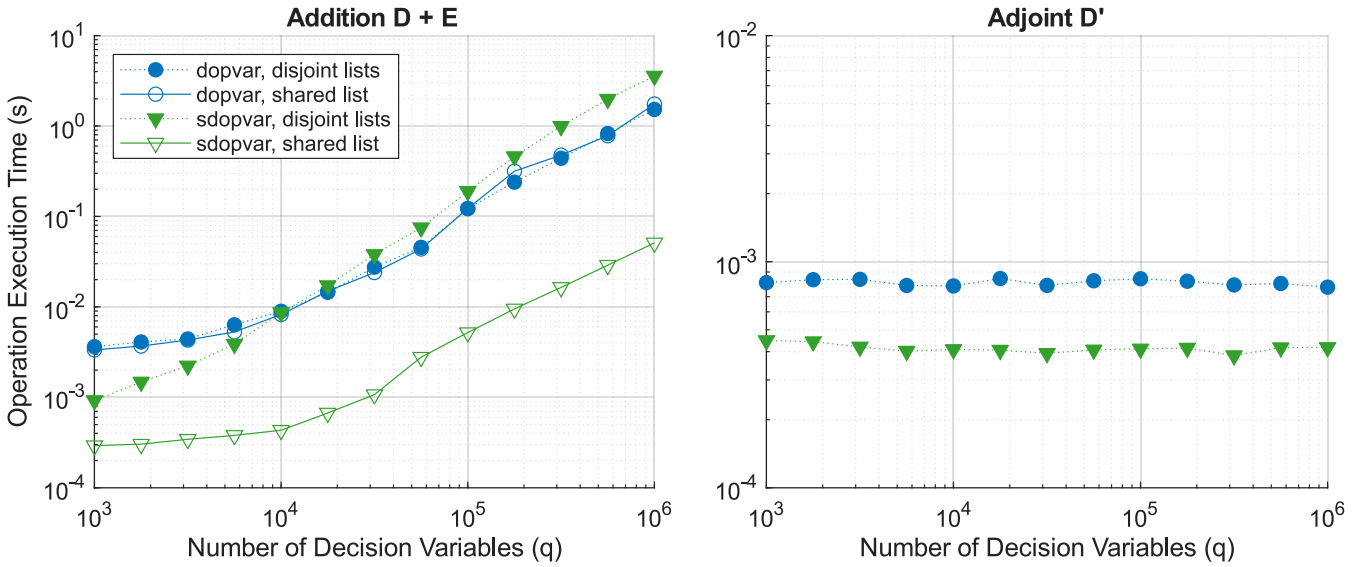


Figure 2. Left: time of the addition $D + E$ of two decision operators with disjoint decision-variable lists (dotted, filled markers) and with one shared list (solid, open markers). Right: time of the adjoint D'. nnz fixed at 110.

5 Parsing the stability LPI against the Lyapunov degree

The analogue of Fig. 4 in Jagt et al. (2022). The LPI is the direct stability test – $P \succ 0$ and $T^{\mathrm{T}}PA + A^{\mathrm{T}}PT \preccurlyeq 0$ – for the heat equation $x_{\mathrm{t}} = \Delta x + \lambda x$ on $[0,1]^{\mathrm{nv}}$ with Dirichlet conditions ($\lambda = \pi^2/2$ in 1-D and π^2 in 2-D, both stable), with the degree d of the Lyapunov variable P swept.

legacy The stock executives `PIETOOLS_PDEstability` and `PIETOOLS_PDEstability_2D` on opvar/dopvar (opvar2d/dopvar2d). The stock executives no longer accept the current `pie_struct`, which holds copvar operators, so the benchmark runs copies with only the input class check removed; the LPI and the SDP are those of the stock executives.

stock SDP The container executive in its stock-slack mode. In 1-D it builds the same SDP as the legacy executive: identical decision variables and PSD block sizes at every d, with 2–13 fewer equality rows because the container layer drops rows that are multiples of rows already written.

sopvar The container executive with its default slack, which builds its own SDP.

Degrees 1-D: every degree of the LPI tied to d, equal to the `veryheavy` settings at $d = 3$. 2-D: the `light` settings with the Lyapunov degrees multiplied by d.

In 1-D the container parses the same SDP in 0.08–0.15 of the legacy time. At $d = 12$ parsing takes 2.5 s against 16.4 s, and its share of the total time falls from 64% to 19%; on that SDP the solver takes 9.1 s (legacy run) and 10.7 s (container run). In 2-D each stack builds its own SDP; the container parses in 0.15–0.46 of the legacy time, and at $d = 3$ the solver takes 96–97% of the total time in both stacks.

Degree 4, parse only. `sossolve` passes no parameters to MOSEK, so a time limit cannot stop a solve, and the 2-D legacy solve at $d = 4$ extrapolates to 1.5–3 h. For such points a stub replaces MOSEK and returns at once with a zero solution; the executive runs to its end, and its whole wall time counts as parsing. At $d = 4$ the container parses its SDP of 3.0 million decision variables in 16.6 s, the legacy stack its SDP of 3.9 million in 56.4 s (ratio 0.29). Where both kinds of run were measured above 3 s, a parse-only time lies within 0.90–1.41 times the parsing time of the full solve: the work outside the LPI build

agrees to 0.23 s, while the build itself varied by 36–47% between calls of one session, for a reason not identified.

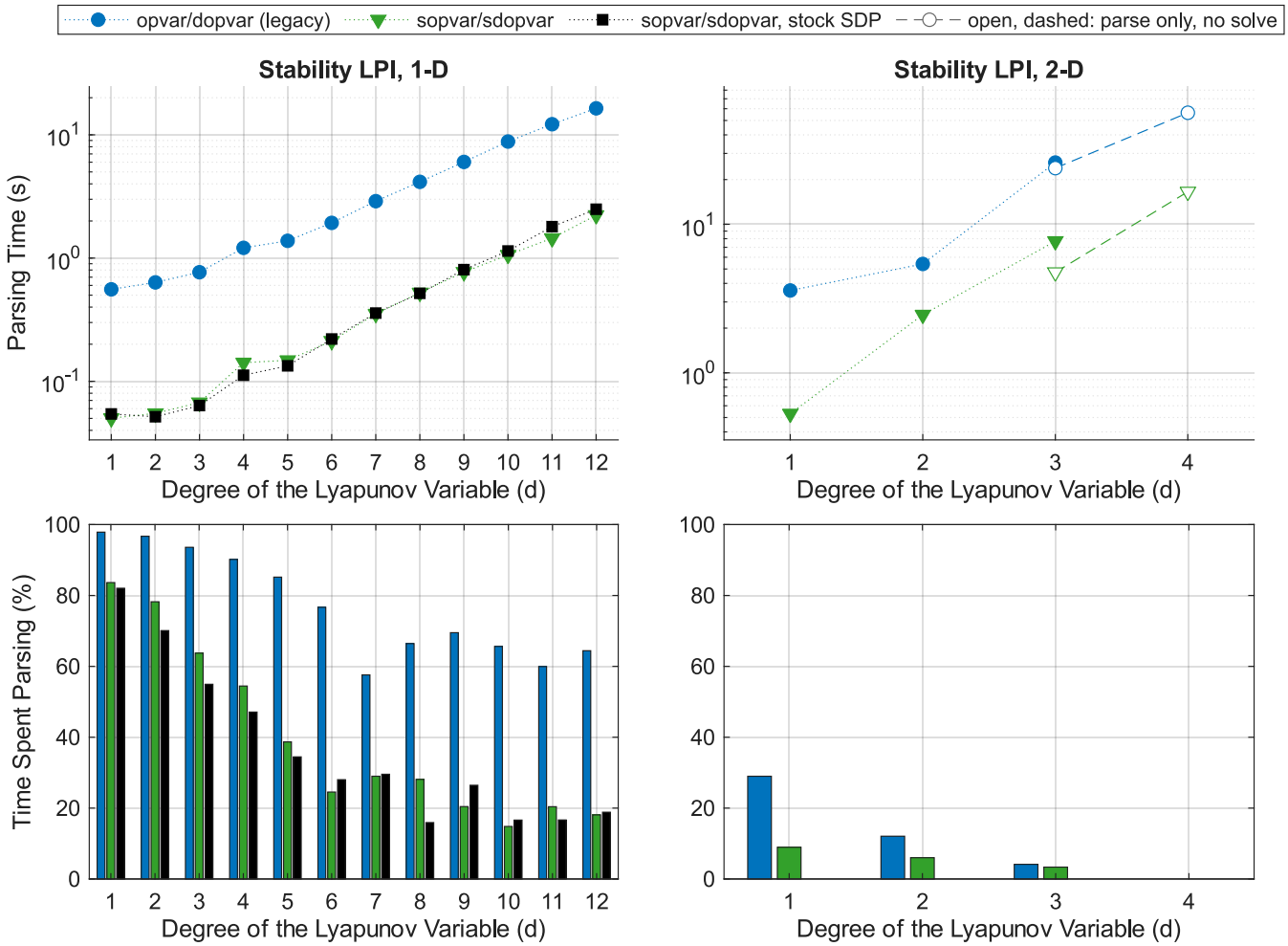


Figure 3. Top: parsing time of the stability LPI against the degree d of the Lyapunov variable, in 1-D (left) and 2-D (right); filled markers on dotted lines for runs with a solve, open markers on dashed lines for parse-only runs. Bottom: percentage of the total time spent parsing, runs with a solve only.

Stability LPI. Parse and solve times in seconds; % = share of total time spent parsing; ndv = decision variables of the SDP. † Parse only.

DIMENSION, D	LEGACY PARSE	STOCK SDP PARSE	SOPVAR PARSE	LEGACY %	STOCK SDP %	SOPVAR %	LEGACY SOLVE	SOPVAR SOLVE	NDV LEGACY	NDV SOPVAR
1-D, 1	0.56	0.054	0.050	98	82	84	0.012	0.010	1,265	1,000
1-D, 4	1.21	0.112	0.142	90	47	54	0.13	0.12	18,254	15,265
1-D, 8	4.16	0.52	0.52	66	16	28	2.10	1.34	132,606	117,641
1-D, 12	16.41	2.49	2.23	64	19	18	9.07	10.09	495,374	453,265
2-D, 1	3.58	–	0.53	29	–	9	8.78	5.38	197,200	187,984
2-D, 2	5.41	–	2.46	12	–	6	39.4	38.5	447,705	594,649
2-D, 3	26.04	–	7.70	4	–	3	604	222	1,992,449	1,455,041
2-D, 4	56.36†	–	16.57†	–	–	–	–	–	3,925,584	3,031,056

6 Parsing the output feedback LPI against the Lyapunov degree

The same measurement for the H∞ dynamic output feedback LPI of `PIETOOLS_Hinf_OFcontrol_sop`, one SDP minimizing the gain bound γ, for the plant $x_t = \Delta x + \lambda x + w_1 + u$ on $[0,1]^{nv}$ with Dirichlet conditions, regulated output $z = [\int x;\, u]$ and measured output $y = \int x + w_2$. With $\lambda = 12$ in 1-D (the plant of DEMO12) and $\lambda = 24$ in 2-D the plant is unstable, since the largest eigenvalue of the Dirichlet Laplacian is $-\pi^2$ on $[0,1]$ and $-2\pi^2$ on $[0,1]^2$. The degrees follow the stability families; the positivity margins are those of DEMO12, 10^{-7}.

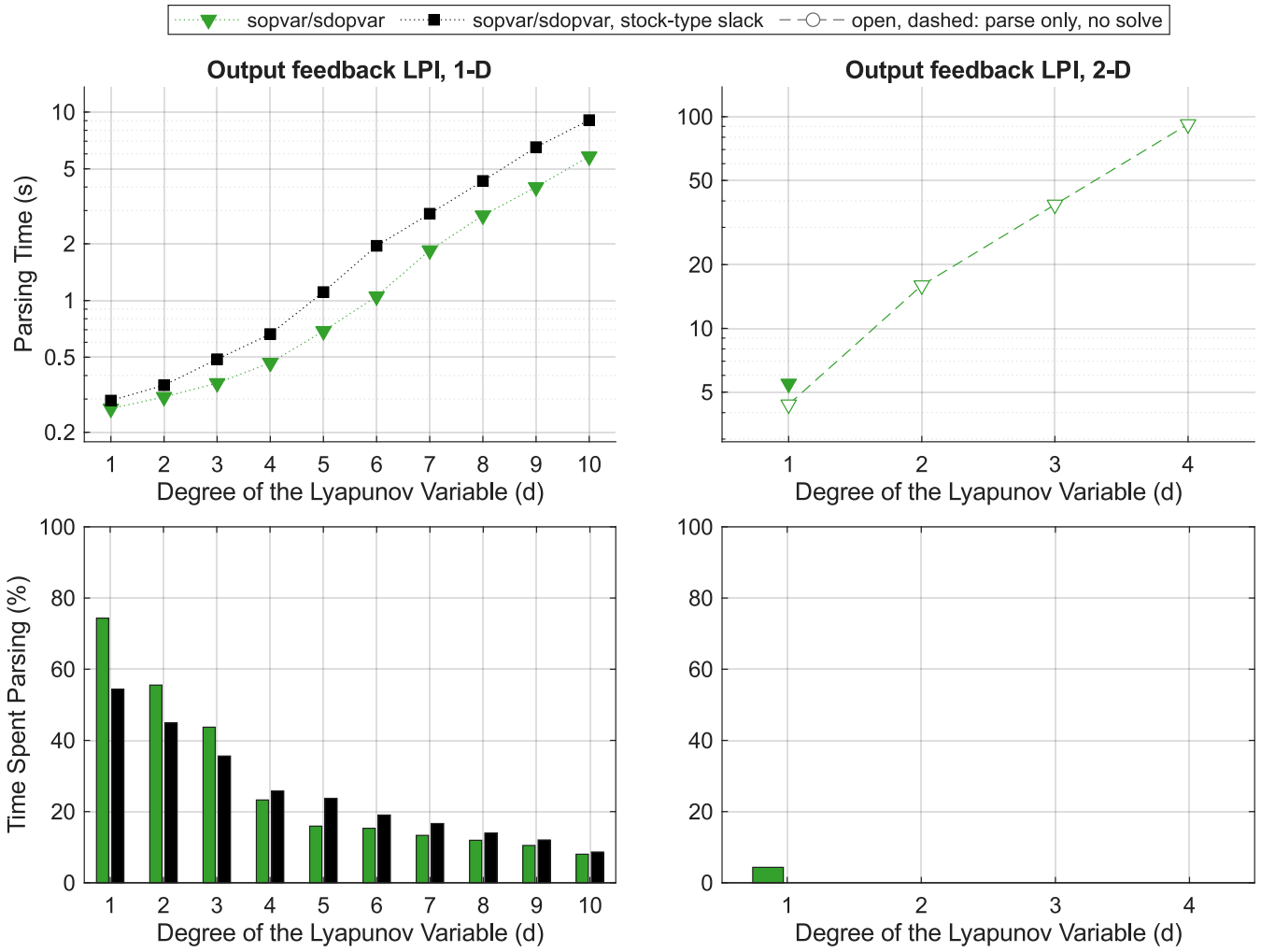


Figure 4. Top: parsing time of the output feedback LPI against the degree d of the Lyapunov variable, in 1-D (left) and 2-D (right); open markers on dashed lines for parse-only runs. Bottom: percentage of the total time spent parsing, runs with a solve only.

stock-type slack The container executive in its stock-slack mode; 1-D only, since in 2-D this mode selects `lpi_ineq_2d`, which has no container counterpart.

sopvar The container executive with its default slack.

No opvar/dopvar implementation of this LPI exists, so the figure has no legacy series. In 1-D ($d = 1$–10, every point solved) the default slack builds 0.54–0.80 of the decision variables of the stock-type slack and parses in 0.54–0.91 of its time; the share of the total time spent parsing falls from 74% to 8%. In 2-D, parse only, the parsing time grows from 4.4 s to 92 s over $d = 1$–4 as the SDP grows from 1.0 to 14.6

million decision variables. At $d = 4$ the output feedback SDP has 4.8 times the decision variables and 7.1 times the equality rows of the 2-D stability SDP, and takes 5.6 times as long to parse. The one 2-D solve, at $d = 1$, takes 121 s and ends with status unknown; parsing is 4% of its total time.

Per decision variable of the SDP, parsing at the top degree of each series takes 4.0–6.3 µs in the container executives – both LPIs, 1-D and 2-D – and 14–33 µs in the legacy executives.

Output feedback LPI. Parse and solve times in seconds; % = share of total time spent parsing; ndv, m = decision variables and equality rows of the sopvar SDP. † Parse only; at 2-D, $d = 1$ the parse-only time is 4.37 s.

DIMENSION, D	STOCK-TYPE PARSE	SOPVAR PARSE	STOCK-TYPE %	SOPVAR %	STOCK-TYPE SOLVE	SOPVAR SOLVE	NDV	M
1-D, 1	0.29	0.27	54	74	0.25	0.089	5,634	623
1-D, 4	0.66	0.47	26	23	1.92	1.52	88,782	2,930
1-D, 8	4.31	2.83	14	12	26.6	20.8	692,278	8,578
1-D, 10	9.06	5.81	9	8	95.8	66.6	1,441,890	12,506
2-D, 1	–	5.48	–	4	–	120.7	1,016,455	18,526
2-D, 2	–	15.97†	–	–	–	–	3,036,545	40,281
2-D, 3	–	38.43†	–	–	–	–	7,184,953	76,199
2-D, 4	–	92.19†	–	–	–	–	14,621,999	131,040

7 Storage

sopvar takes 0.45–0.93 of the opvar storage for $n, d \le 8$, and sdopvar 0.33–0.64 of the dopvar storage. For large q the decision-variable names dominate both: at $q = 10^6$ with 168 nonzero decision coefficients ($n = d = 2$), the sdopvar occupies 133.8 MB – 134 bytes per name for its one list – and the dopvar 401.3 MB, one list in each of its three kernels. MATLAB `whos` counts a list shared copy-on-write once per holder, so 401.3 MB bounds the dopvar from above.

8 Limits

- **One spatial dimension.** opvar and dopvar are 1-D classes, so the class comparison is 1-D; opvar2d and dopvar2d enter only through the 2-D LPI.
- **Single blocks.** Operators with $\mathbb{R}^m$ components are containers (copvar, cdopvar). `sopvar/Testfolder/Bench_copvar_vs_legacy.m` (09/18/2026) compares those with opvar for addition, adjoint and composition only.
- **Synthetic structure.** Kernels are dense random polynomials and the decision structures are synthetic; `lpivar` and `poslpivar` produce other sparsity patterns.
- **Solver status.** The stability solves in 1-D at $d = 9$ (legacy and stock SDP), $d = 10$ (legacy) and $d = 11$ (sopvar) end with status unknown, as do the output feedback solves at 1-D, $d = 8$ (stock-type slack) and 2-D, $d = 1$; parsing time does not depend on the status.
- **2-D.** The two stacks build different SDPs, so the 2-D solve times compare formulations, not data structures. Solves stop at $d = 3$ for stability, where one legacy solve takes 604 s, and at $d = 1$ for output feedback; the larger degrees are parse only.
- **Parse-only points.** Within 0.90–1.41 times the full-solve parsing time where both were measured above 3 s. The build of one LPI varied by 36–47% between calls of one session, so two single points closer than that are not distinguished by these measurements.

- **First calls.** The 2-D stability solves at $d = 3$ ran as the first calls of a fresh MATLAB session and carry its one-time cost; at that degree the first parse-only call of sopvar takes 6.94 s, the next two 4.75 s and 4.54 s. The 2-D output feedback solve at $d = 1$ parses in 5.48 s warm and in 11.89 s as the first call of a session.
- **Timing.** One machine, medians of warm calls. In a fresh MATLAB session the first call carries a one-time cost (1-D, $d = 1$: 0.89 s legacy and 0.23 s container, against 0.56 s and 0.05 s warm); a discarded warm-up pass removes it. At $d = 8$ the first call at a new degree lies within 20% of the repeats. Storage is `whos` of the result, not peak memory.

9 Data

All data and scripts are in `baselines/2026-11-05_class_ops` on the ndopvar branch (commits 8343fd1b, 7688dcf2 and 392a6adf): `results.csv` (operations at fixed size), `scaling/ops_vs_q.csv` (every operation against q), `scaling/stability_parse_vs_degree.csv` and `scaling/of_parse_vs_degree.csv` (parsing and solve times against d, with the parsing time of every call and a parse-only flag), the figures as PNG and PDF, and the scripts in `private/`, the stub solver in `private/stub_mosek/`. `scaling/README.md` lists the commands that regenerate everything; the benchmark needs the opvar classes, so the CSV files remain the record once those classes are retired.

D. Jagt, S. Shivakumar, P. Seiler and M. M. Peet, “Efficient data structures for representation of polynomial optimization problems: implementation in SOSTOOLS,” *IEEE Control Systems Letters*, vol. 6, pp. 3493–3498, 2022.

Generated from the committed data in `baselines/2026-11-05_class_ops`.